École Nationale Supérieure d'Informatique
2CS SIQ

Appendices

Supplementary Technical Material

Detailed protocol references, VNI/subnet mappings, QoS configurations, and security validation procedures.

Contents

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A Protocol Stack: Fabric Inventory and Addressing

A.1 Spine and Leaf Loopback Assignments

Node	Loopback0	ASN	Role
S01	10.1.0.1/32	65000	Spine / EVPN distribution
S02	10.1.0.2/32	65000	Spine / EVPN distribution
L01	10.1.0.11/32	65001	Edge Pod A
L02	10.1.0.12/32	65001	Edge Pod B
L03	10.1.0.13/32	65002	Core Pod A
L04	10.1.0.14/32	65002	Core Pod B
L05	10.1.0.15/32	65003	Storage Pod A
L06	10.1.0.16/32	65003	Storage Pod B
L07	10.1.0.17/32	65004	Compute Pod A
L08	10.1.0.18/32	65004	Compute Pod B
L09	10.1.0.19/32	65005	Ultra Compute Pod A
L10	10.1.0.20/32	65005	Ultra Compute Pod B

A.2 P2P Underlay Addressing

Only one /31 is allocated per Spine-to-Leaf link. The underlay uses two small, contiguous pools:

Pool	Range	Usage
S01 links	10.0.0.0–18 (/31)	One /31 per S01-to-Leaf uplink across all five Leaf pairs
S02 links	10.0.1.0–18 (/31)	One /31 per S02-to-Leaf uplink across all five Leaf pairs

A.3 EVPN Route Type Usage

Type	Function	Fabric usage
Type 1	Ethernet Segment	ESI multihoming state and designated forwarder election
Type 2	MAC/IP route	MAC/IP bindings and ARP suppression within domains
Type 3	IMET	Head-end replication lists per VNI
Type 5	IP prefix	Intra-domain routing and cross-domain imports

A.4 BGP and Control Plane Details

- **Underlay model:** eBGP advertises only loopback /32s; P2P /31s are configured as link-local and not advertised.
- **EVPN control:** MP-BGP with ‘address-family l2vpn evpn‘ on all eBGP sessions.
- **Next-hop preservation:** Spines use ‘next-hop-unchanged‘ for the EVPN address family so VXLAN tunnels terminate directly on Leaf VTEPs, not on Spines.
- **Server multihoming:** Servers dual-home with LACP (Bond or LAG) and EVPN ESI (Ethernet Segment Identifier); no MLAG peer-link between Leaf pairs.
- **Symmetric IRB:** Each Leaf performs first-hop routing for the VNIs it locally serves. Inter-subnet forwarding uses symmetric IRB with per-domain L3VNIs and fabric-wide anycast gateways.
- **BUM handling:** Unknown unicast and multicast are replicated by ingress Leaf using head-end replication (HER). No PIM is used in the overlay.
- **BFD convergence:** BFD is enabled on all BGP sessions for sub-second failure detection and route withdrawal. Timers: min-rx 100 ms / min-tx 100 ms / multiplier 3 (300 ms detection window). Total convergence including BGP withdrawal and RIB/FIB re-installation is targeted at under 1 second on non-congested hardware; this is a design target, not a hard guarantee.

A.5 Fabric MTU and Performance

- **Link speeds:** 100G between Spine and Edge Pod plus Spine and Ultra Compute Pod; 25G for standard Leaf uplinks; 10G for border router attachment. The 10G ISP attachment creates a 10:1 asymmetry relative to Ultra Compute Pod’s 100G internal bandwidth; this is accepted given that GPU/HPC workloads are primarily intra-fabric. External transfer demand should be reviewed if off-site data volumes increase significantly.
- **Fabric MTU:** 9000 bytes on all fabric links to accommodate VXLAN and Ceph headers without fragmentation.
- **Host MTU:** VXLAN adds approximately 50 bytes of outer header overhead. Hosts configured with jumbo frames must use a maximum MTU of 8950 bytes (9000 – 50) to prevent outer-frame overflow. Storage Pod (Ceph) and

Ultra Compute Pod (RoCEv2) hosts are configured at 8950 bytes. All other pods use standard 1500-byte host MTU, which is unaffected by VXLAN overhead.

- **Server connectivity:** 25G by default; 100G for Ultra Compute Pod servers; 10G for border router and OOB management.
- **RoCEv2 and QoS:** RoCEv2, PFC, and ECN are scoped to the Ultra Compute Pod (L09/L10) to avoid fabric-wide side effects.

B VNI and Subnet Segment Mappings

B.1 Complete VNI/VLAN/Subnet Table

The following table is the authoritative reference for all logical network segments in the fabric, including their VNI assignments, VLAN identifiers, communication domain membership, host subnets, and Leaf pair attachment. VNI values follow the convention 10 000 + VLAN ID.

VNI	VLAN	Segment	Comm. Domain	Subnet	Leaf	Notes
10010	10	STUDENT-TP	Pedagogy	192.168.10.0/24	L07/L08	Supervised lab servers (TP)
10020	20	STUDENT-PROJ	Pedagogy	192.168.20.0/24	L07/L08	Student project VMs
10030	30	LMS-STAFF	Staff	192.168.30.0/24	L03/L04	Moodle LMS instance (student access via firewall policy from the Pedagogy domain)
10040	40	SERVICES-WEB	Staff	192.168.40.0/24	L03/L04	Internal web portals
10050	50	CORE-INFRA	Staff	192.168.50.0/24	L03/L04	OpenLDAP, DNS/DHCP, monitoring aggregation, automation control plane; also hosts JupyterHub, its state database, Munge, and SLURM control/accounting plane
10060	60	HR-FINANCE	Administration	192.168.60.0/24	L03/L04	HR and Finance applications with encrypted storage volumes
10070	70	AI-GPU	Staff	192.168.70.0/24	L09/L10	GPU nodes; RoCEv2, PFC, and ECN isolated to this pod only
10080	80	STORAGE-SAN	Staff	192.168.80.0/24	L05/L06	Storage backend: Ceph cluster nodes, NAS heads, and backup appliances
10090	90	BAC-ORIENT	Orientation	192.168.90.0/24	L01/L02	Seasonal BAC Orientation servers; third ISP fiber only; zero production routes
10100	100	DMZ-WEB	Public	192.168.100.0/24	L01/L02	DMZ web servers; no RFC1918 routes imported; isolated by routing table structure
N/A	110	MGMT-OOB	OOB (not EVPN)	172.16.0.0/24	—	Informational only: out-of-band management on dedicated OOB switch; no EVPN VNI assignment
10120	120	WIFI-CTRL-MGMT	Wireless Control	192.168.120.0/30 + static route 192.168.120.10/32	L01	WiFi controller management; local VRF on L01 only; connected link subnet plus single static /32 controller route; no fabric-wide L3VNI; no default route

B.2 Design Rationale

B.2.1 Pedagogy Communication Domain (VNI 10010–10020)

Student workloads are placed in the Compute Pod (L07/L08) where oversubscription and buffering are tuned for bursty, latency-tolerant traffic. Connectivity to Staff communication-domain services is explicit and filtered through the Edge firewall.

B.2.2 Staff Communication Domain (VNI 10030–10080)

Production services, shared storage, and compute control planes are grouped in the Staff communication domain. The LMS and Services segments are attached to the Core Pod (L03/L04) for infrastructure affinity. AI/GPU workloads are in the Ultra Compute Pod (L09/L10) with 1:1 oversubscription and loss-sensitive transport controls. Storage is attached to the Storage Pod (L05/L06) to avoid distorting Core Pod latency characteristics.

B.2.3 Administration Communication Domain (VNI 10060)

HR/Finance systems are isolated in the Administration communication domain with tighter access controls and mandatory encryption for sensitive data. Paths to and from other communication domains are explicit firewall-mediated permits only.

B.2.4 Orientation Communication Domain (VNI 10090)

The BAC Orientation platform is deliberately isolated with zero routes to production. It is powered down outside the active season and connected only to a dedicated third-party ISP fiber.

B.2.5 Public Communication Domain (VNI 10100)

DMZ web services are in the Public communication domain with no imported internal routes. This is a routing table property, not a firewall policy: a compromised DMZ host has no routing path to internal networks.

B.2.6 Management Networks (MGMT-OOB, VNI 10120)

The OOB management segment is not part of EVPN and has no VNI assignment; it is a flat Layer-2 network on a dedicated switch with no general L3 path into the production fabric. The only allowed OOB exception is TACACS+ LDAP queries to the dual-homed OpenLDAP service on TCP/389. The WiFi controller management path uses a local VRF on L01 only, with connected subnet 192.168.120.0/30 and static route 192.168.120.10/32, and is not a fabric-wide communication domain.

C Quality of Service Architecture

QoS in this fabric is pod-aware rather than attempting to equalize all traffic. Control-plane traffic is protected first, the Ultra Compute Pod gets lossless-capable treatment with RoCEv2: ECN combined with DCQCN (Data Center Quantized Congestion Notification) is the primary congestion management mechanism, with PFC (Priority Flow Control, 802.1Qbb) enabled only where RDMA-capable NICs require it and cannot operate with ECN-based feedback alone. This order of preference limits head-of-line blocking propagation to only those queues where PFC is strictly necessary. Critical seasonal services are protected during their active window, and student plus bulk flows consume residual capacity. This model prevents a single traffic class from degrading infrastructure stability or performance-sensitive workloads.

C.1 Traffic Classification and Priority

Prio	Class	Marking and Treatment	Examples
1	Network Control	CS6; strict priority	eBGP, EVPN transport, BFD, LACP
2	Real-Time	EF; strict priority, rate-limited	Reserved for future voice/video services
3	Critical Applications	AF41; WFQ high; 20% min	BAC Orientation, authentication, admin flows
4	HPC / AI	AF31; WFQ high; 25% min	AI-GPU (VNI 10070), SLURM/K8s, latency-sensitive I/O
5	Academic Apps	AF21; WFQ medium; 20% min	LMS, web portals, core services, supervised labs
6	General Student	AF11; WFQ low; 10% min	Student projects and general browsing
7	Bulk / Backup	CS1; scavenger; 5% min	Backup replication, image sync, software updates
8	Best Effort	DF; residual bandwidth	Unclassified traffic and overflow

C.2 VNI to DSCP Mapping

Segment identity informs QoS, but does not define it completely: `STORAGE-SAN` can carry either AF31 or CS1 traffic depending on whether the flow serves latency-sensitive GPU work or bulk replication and backup.

DSCP	Segments	Class
CS6	eBGP, EVPN, BFD, LACP (control plane)	Network Control (strict priority)
AF41	BAC-ORIENT (10090)	Critical Applications
AF31	AI-GPU (10070), latency-sensitive STORAGE-SAN (10080) flows serving GPU jobs	HPC / AI (high, lossless where capable)
AF21	LMS-STAFF (10030), SERVICES-WEB (10040), CORE-INFRA (10050), STUDENT-TP (10010)	Interactive / Academic
AF11	STUDENT-PROJ (10020)	General Student
CS1	Backup, image distribution, software updates	Bulk / Backup (scavenger)
DF	All other traffic	Best Effort

C.3 Enforcement and Implementation

- **Ingress classification:** Leaves classify traffic by ingress interface, VNI, and service role, setting DSCP before forwarding into the fabric.
- **Transit scheduling:** Spines schedule already-marked traffic into priority queues over equal-cost transit paths. No re-marking occurs in transit.
- **Egress shaping:** Edge Pod shapes north-south egress to ISP committed rates and polices untrusted inbound bursts; external DSCP is re-marked upon ingress.
- **RoCEv2 and congestion control:** ECN combined with DCQCN is the primary congestion management mechanism for AF31 traffic on Ultra Compute uplinks (L09/L10 to Spines). PFC is enabled only where RDMA-capable NICs require lossless delivery and cannot operate with ECN-based signalling alone, limiting pause-frame propagation to the minimum necessary scope.
- **OOB isolation:** Out-of-band management traffic remains outside production QoS policies.
- **Observability:** Per-class counters, queue drops, and ECN marks are exported to the monitoring stack, and the policy is deployed through automation.

C.4 Rationale

- **Pod-aware model:** QoS follows workload role and service criticality rather than segment identity alone. Ultra Compute gets lossless-capable treatment because GPU/HPC workloads require it; the Compute Pod allows oversubscription and burst absorption.
- **Structural protection:** Network control traffic (CS6) is always strict priority so fabric convergence is not impaired by user traffic congestion.
- **Scoped loss-sensitivity:** RoCEv2 transport uses ECN/DCQCN as the primary congestion signal and PFC only as a last-resort lossless guarantee, both confined to Ultra Compute Pod fabric links. This avoids fabric-wide head-of-line blocking and unnecessary complexity on pods with tolerant traffic.
- **Seasonal flexibility:** BAC Orientation traffic gets AF41 priority only during its active window; outside that window, the domain is powered off.

D Security Validation and Testing

D.1 Validation Framework

The security architecture is validated against a comprehensive threat model and a set of structural tests. Validation occurs at provisioning, during major topology changes, and as part of the annual operational review. The following tests ensure that structural isolation, control-plane hardening, and policy enforcement remain effective.

D.2 Validation Test Suite

Validation	Target	Pass condition
Domain isolation	VRF boundaries	Unauthorized pairs have 100% packet loss
VXLAN injection	VXLAN ACL	Crafted server traffic is dropped silently
BGP auth	TCP MD5	Session drops with wrong key and recovers on fix
Max-prefix	External peers	Session tears down when peer advertises more than 4 prefixes
IDS alerting	Edge mirror feed	Scan or exploit signatures alert on the IDS node
Bastion enforcement	SSH path	No direct SSH banner from non-Bastion IPs
nftables lateral	Host firewall	Blocked lateral traffic is refused or times out
OOB isolation	Mgmt network	No production reachability from OOB segment
BFD failover	Spine uplink loss	Reconvergence under 1 second (baseline: min-rx/min-tx 100 ms, multiplier 3)
ESI failover	One Leaf loss	Bond stays up and traffic continues
Public domain	DMZ to internal	100% loss on packets to internal RFC1918 subnets
Secure boot (BAC)	Orientation nodes	Alternate boot media is blocked or requires physical override
Syslog correlation	Central logging	Security events arrive with synchronized timestamps

D.3 Threat Mapping and Controls

The security architecture addresses the following threat categories:

ID	Threat	Primary mitigation
T1	External intrusion	Edge Pod stateful firewall
T2	BGP route injection	TCP MD5, route filtering
T3	BGP table exhaustion	Max-prefix guard on external peers
T4	VXLAN frame injection	VXLAN source ACL at Leaves
T5	East-west lateral movement	Intra-domain routing isolation
T6	Cross-domain movement	VRF boundaries, Edge firewall
T7	Unauthorized admin access	Bastion-only SSH, key-based auth
T8	BAC data tampering	Seasonal disconnect, secure boot
T9	Insider or rogue admin	Bastion control, AWX-managed change discipline, central logging
T10	Undetected intrusion	Suricata IDS and syslog retention
T11	Physical access	Locked equipment, TPM secure boot
T12	Clock desynchronization	Chrony NTP with sub-1s skew target

D.4 Control Layers

D.4.1 Ring 1: Perimeter Firewall

HA firewall pair in Edge Pod with stateful Layer 7 inspection. All north-south and authorized inter-domain traffic is inspected and logged there.

D.4.2 Ring 2: BGP Hardening

TCP MD5 signature on all BGP sessions. External ISP peers accept inbound default route only, RFC1918 space is blocked from outbound advertisement, and max-prefix guard tears down an external session if the route count exceeds 4 prefixes per peer. *TCP MD5 note:* TCP MD5 uses MD5-based XOR obfuscation (RFC 2385) and is retained for broad NOS compatibility. TCP Authentication Option (TCP-AO, RFC 5925) is the cryptographically stronger successor and should be adopted when all fabric devices confirm support.

D.4.3 Ring 3: Control-Plane ACLs

BGP (TCP/179) and BFD (UDP/3784) are permitted only from fabric loopback addresses. VXLAN (UDP/4789) ingress is permitted only from known Leaf VTEP addresses. SSH is permitted only from the Bastion host (172.16.0.50). Management ports on switches are not directly accessible from production segments.

D.4.4 Ring 4: Bastion Access

All administrative access to switches and servers transits through the Bastion host. SSH uses ed25519 keys only; password authentication is disabled. ‘AllowUsers‘ whitelist restricts access to authorized operators. ‘MaxAuthTries 3‘ and ‘LoginGraceTime 30‘ prevent brute-force attacks. Failed attempts are logged and correlated.

D.4.5 Ring 5: Host Micro-Segmentation

Every server runs host-level firewall (‘nftables‘ or equivalent). Default ‘INPUT‘ policy is ‘DROP‘. ‘ESTABLISHED‘ and ‘RELATED‘ traffic is allowed. SSH inbound is permitted only from the Bastion. Role-specific services are opened explicitly.

D.4.6 Supporting Controls

- **IDS:** Suricata runs on a dedicated IDS host fed by a mirror of all Edge Pod traffic.
- **Logging:** Syslog is retained for 90 days locally plus 1 year in S3-compatible RGW storage.
- **NTP:** Chrony maintains clock synchronization within 1 second across all nodes for forensics and authentication auditing.
- **OOB isolation:** The OOB management network (172.16.0.0/24) has no general routes into production EVPN domains; the only allowed exception is TACACS+ LDAP queries to OpenLDAP on TCP/389.
- **IDS management:** IDS host management address (172.16.0.51) is confined to OOB and isolated from production traffic.

D.4.7 BAC-Specific Controls

Orientation platform is powered off approximately 10 months per year. When activated:

- All Orientation servers boot with TPM-backed secure boot to prevent unauthorized OS modification.
- Physical power and boot controls are locked to prevent unauthorized restarts.
- Orientation domain has zero routes to production; it can only reach external networks via a dedicated third-party ISP fiber.
- A single management IP range is permitted inbound only during the active operational window.

D.5 Monitoring and Alerting

- **BGP session monitoring:** Alerts fire on session state changes, route count spikes, or unexpected next-hop changes.
- **Domain isolation monitoring:** Periodic cross-domain connectivity tests verify that unauthorized routes have not been inadvertently created.
- **Firewall flow logging:** All denied flows are logged. Suspicious patterns (port scanning, DoS indicators) trigger alerts.
- **SSH access logging:** All authentication attempts are logged. Failed attempts are correlated across Bastion, switches, and servers.
- **Link flapping:** Rapid interface state transitions trigger investigation and potential circuit diagnostics.